Conservation Policy with Political Resistance

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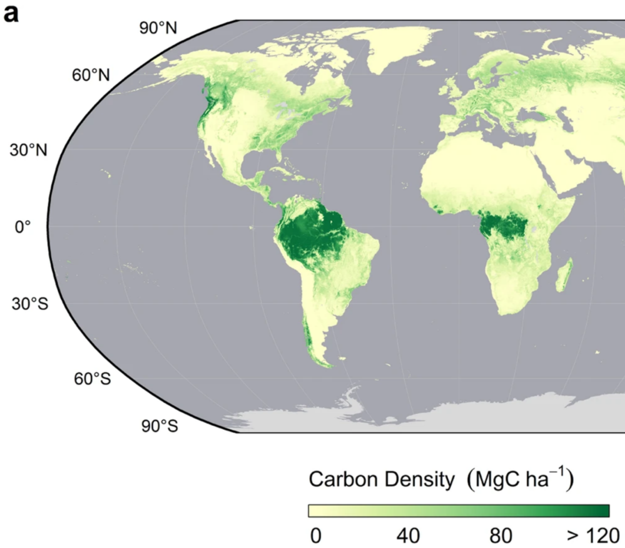
Forest conservation protects major carbon stocks



(Spawn et al. 2020)

But conservation is political

New Internationalist

View from Brazil: Agribusiness lobby scuppers climate gains

Lula wants Brazil to be a beacon in the fight against global warming, Leonardo Sakamoto. The powerful lobby that represents agribusiness in...

Sep 4, 2023

Mongabay

Indonesia palm oil lobby pushes 1 million hectares of new Sulawesi plantations

A state-owned palm oil company and an industry association have begun early work to push a vast new plantation strategy in Sulawesi,...

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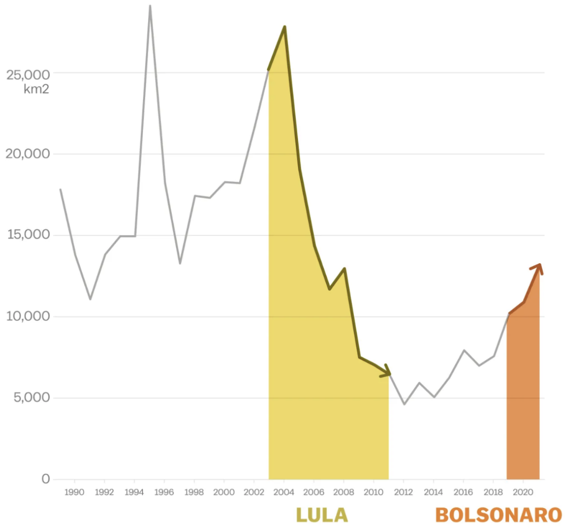
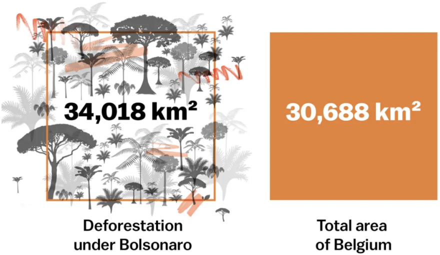
And politics threatens conservation





(Northrop 2022)

Question

How can environmental regulation navigate political resistance?

This paper

1. Regulation creates political resistance
   - In Brazil, forest conservation mobilized agricultural interests
2. Political resistance shapes optimal policy
   - Producer burden raises repeal risk, reducing feasible welfare gains
3. Targeting preserves environmental gains
   - Prioritize locations with high emissions and low agricultural value

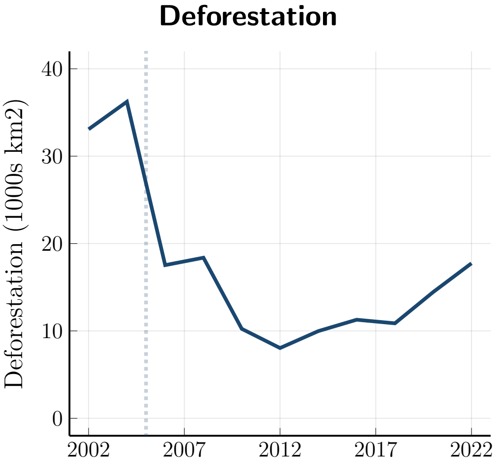
Regulation creates political resistance

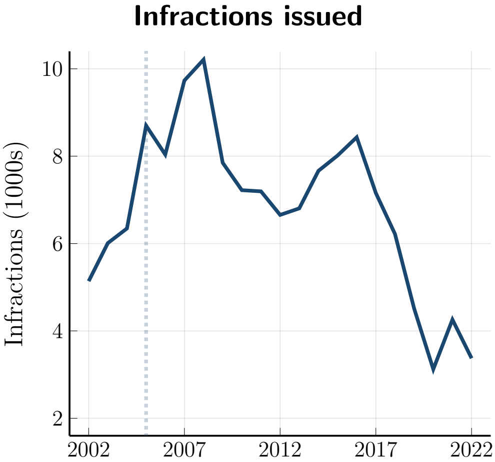
The Brazilian Amazon

- **PPCDAm** strengthened forest protections beginning in 2004
  - Forest Code requires 80% of each property to remain forested
  - Satellite monitoring, protected areas, and tougher penalties

- Deforestation fell sharply through 2012
  - And then rose again as enforcement weakened

PPCDAm strengthened enforcement



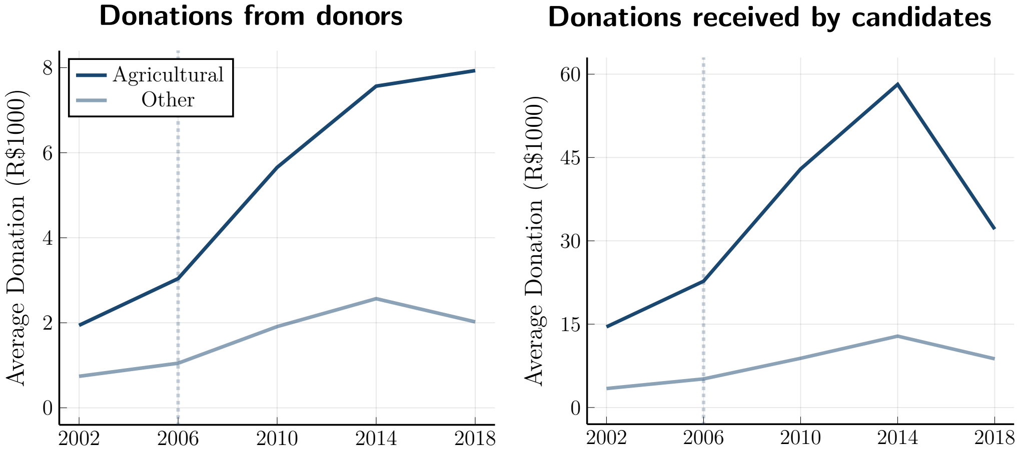


Measuring political resistance

	Donors	Candidates
Full sample	7,337	980
Agricultural firm owners	810	134
Environmental infractors	322	166

- State and federal legislative elections from 2002 to 2018
  - Universe of formal campaign donations (TSE)
  - Linked to firm ownership (*Receita Federal*)
  - And to environmental infractions (IBAMA)

Agricultural donations rose after PPCDAm

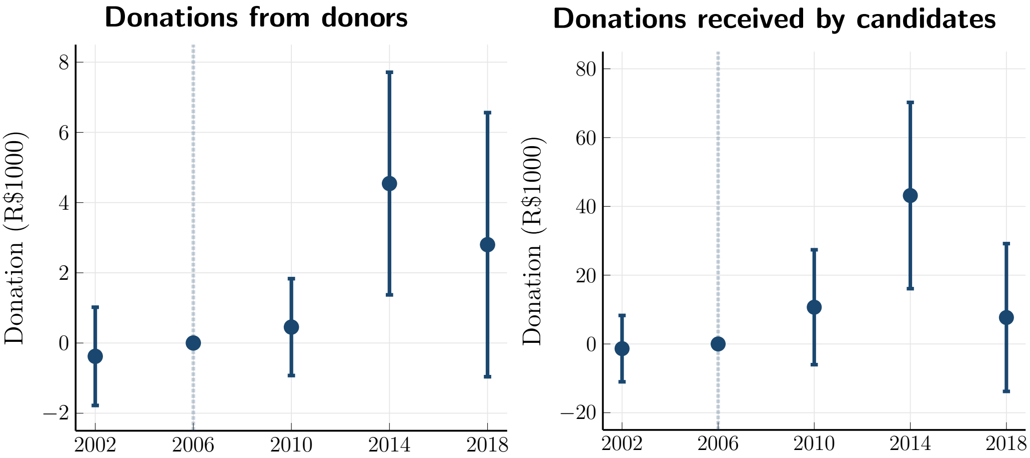


Difference-in-differences

$$y_{imt} = \sum_s \beta_s \, \text{Agricultural}_i \times \mathbb{1}\{t = s\} + \alpha_i + \gamma_{mt} + \varepsilon_{imt}$$

- Agricultural vs. other firm owners, before vs. after PPCDAm
- Individuals i, municipalities m, election years t
  - Municipality-by-year and individual fixed effects
  - Matching on firm age, structure, size, capital, tax regime (entropy balancing)

PPCDAm mobilized agricultural donations



Agricultural donors doubled their giving

	All candidates	Agricultural candidates
$Ag \times Post$ 2006	2.79***	2.33***
	(1.02)	(0.62)
Effect as % of mean	116	212
Observations	35,195	35,195
Fixed effects	$i + mt$	$i + mt$
Entropy balancing	x	x

Especially where enforcement was binding

Standardized H:	Deforestation reduction	Cloud coverage
$Ag \times Post\ 2006$	1.50**	7.41***
	(0.76)	(2.58)
$Ag \times Post\ 2006 \times H$	1.82**	-11.78**
	(0.89)	(4.72)
Observations	33,615	27,385
Fixed effects	$i + mt$	$i + mt$
Entropy balancing	x	x

Agricultural candidates received more donations

	All donors	Agricultural donors
$Ag \times Post$ 2006	21.19***	6.58***
	(7.07)	(2.12)
Effect as % of mean	95	132
Observations	3,480	3,480
Fixed effects	$i + mt$	$i + mt$
Entropy balancing	x	x

And more votes (!)

	All elections	Federal elections	State elections
$Ag \times Post$ 2006	10.05***	21.49**	1.95*
	(2.82)	(8.47)	(1.12)
Effect as % of mean	46	47	16
Observations	1,328	260	821
Fixed effects	$i + mt$	$i + mt$	$i + mt$
Entropy balancing	x	x	x

Measuring agricultural value

$$\overline{PS}_m \propto \frac{L_m}{N_m} \log\left(1 + \frac{A_m}{F_m}\right), \qquad \overline{R}_m = \frac{R_m}{N_m}$$

1. **Implied surplus** via logit model (observed choices reveal underlying value)
2. **Observed revenue** via census data (model-free but misses economic costs)

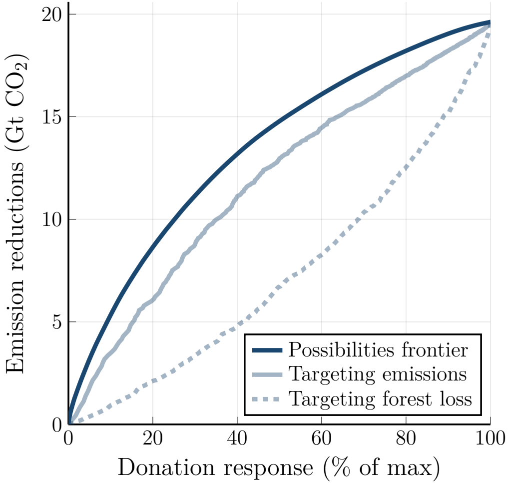
- Municipality-level data: pre-period, per producer, logged, standardized

Political resistance rises with agricultural value

Standardized H:	Implied surplus	Observed revenue
$Ag \times Post$ 2006	1.83**	1.83
	(0.91)	(1.12)
$Ag \times Post$ 2006 $\times H$	2.75***	2.17*
	(0.93)	(1.11)
Observations	33,880	33,880
Fixed effects	$i + mt$	$i + mt$
Entropy balancing	x	x

The political possibilities frontier

- Predicted abatement vs. resistance
  - Each step adds a municipal ban
- Targeting emissions
  - Ignores political resistance
  - Bans where emissions are highest
- Targeting forest loss
  - As priority lists do in practice
  - Bans where resistance is highest (!)



Political resistance shapes optimal policy

The Pigouvian benchmark

$$a_i(\tau_i) = \mathbb{1}[v_i > \tau_i], \qquad w_i(\tau_i) = (v_i - \kappa e_i)a_i(\tau_i)$$

- Plot i generates emissions e_i and agricultural value $v_i > 0$
    - Emissions have social cost κ
- Pigouvian tax maximizes welfare ($\tau_i^P = \kappa e_i$)
    - But also imposes substantial producer burden
    - Tax revenue is a transfer from producers to government

Abatement and burden can diverge

$$b_i(\tau_i) = PS_i(0) - PS_i(\tau_i) = \min\{v_i, \tau_i\}$$

	Value	Production	Abatement	Burden
Marginal	$v_i < \tau_i$	Stops	e_i	v_i
Inframarginal	$v_i > \tau_i$	Continues	0	τ_i

Political resistance changes the objective

$$F(\tau) = \underbrace{[1 - R(B(\tau))]\, W(\tau)}_{\text{regulation survives}} + \underbrace{R(B(\tau))\, W(0)}_{\text{regulation repealed}}$$

- Resistance rises with total producer burden $B(\tau)$
  - Raises repeal risk with $R'(B) > 0$
- Optimal policy maximizes **feasible welfare** $F(\tau)$
  - Balances welfare gains against repeal risk

Optimal policy targets high-emission, low-value plots

$$\tau_i^* = \kappa e_i \cdot \mathbb{1}\left[\eta_i < \frac{1}{1+\mu^*}\right], \qquad \eta_i = \frac{v_i}{\kappa e_i}$$

- Pigouvian with a cutoff
  - Tax at $\tau_i^P = \kappa e_i$ only for low η_i
  - Exempt all efficient $(\eta_i > 1)$ and some inefficient
- Political resistance tightens the cutoff
  - By raising shadow cost μ^*

Targeting can preserve environmental gains

Can practical policies realize the gains from targeting?

- Optimal policy targets emissions and agricultural value
- Practical policies use less information
  - Emissions are observable via satellites
  - But agricultural values are private information
- We use an empirical model to infer agricultural values
  - Then compare optimal and practical policies

Agricultural land use

$$\max\left\{v_i - \tau_i + \epsilon_{ij}^a,\ \epsilon_{ij}^f\right\}, \qquad \log\left(\frac{a_i}{f_i}\right) = \frac{v_i - \tau_i}{\sigma}$$

- Site i chooses agriculture or forest for each hectare j
  - Given agricultural value v_i and regulation τ_i
  - Logit shocks $(\epsilon_{ij}^a, \epsilon_{ij}^f)$ have scale σ
- Land use shares reveal agricultural value relative to σ
  - The model predicts land use and producer surplus under any τ_i

Logit estimation

$$\log\left(\frac{a_{it}}{f_{it}}\right) = \beta r_{it} + \alpha_i + \gamma_t + \varepsilon_{it}, \qquad \sigma = \frac{1}{\beta}$$

- Spatial panel of 0.25-degree cells from 2001 to 2019
  - Land use, crop suitability, market access, carbon stocks
  - Returns combine suitability with market prices (soy, maize, pasture)

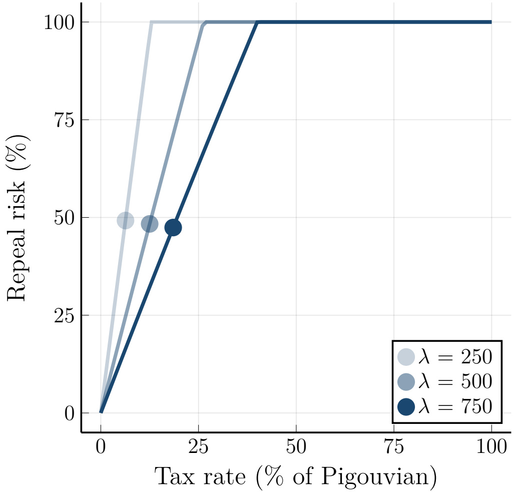
- Pooled 10-year long differences capture gradual frontier adjustment
  - Nine overlapping windows w: 2001–2011 to 2009–2019

Estimates of β

	Crop weights for returns		
	Baseline shares	Leave-one-out shares	Fitted shares
Agricultural returns	0.126***	0.134***	0.128***
	(0.045)	(0.048)	(0.049)
Observations	46,682	46,682	46,682
Fixed effects	$iw + tw$	$iw + tw$	$iw + tw$

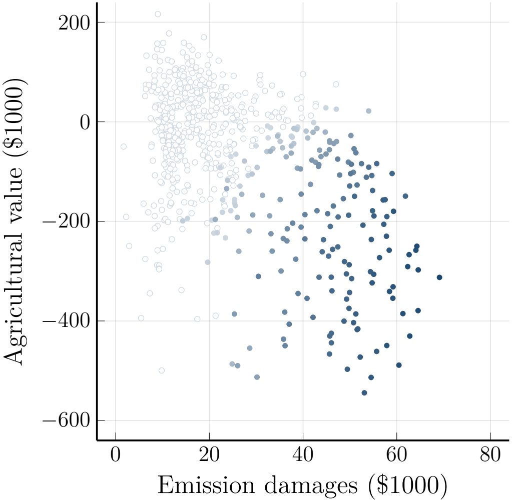
Optimal policy depends on repeal risk

- The land use model gives $B(\tau)$
- We specify repeal risk $R(B)$
  - Repeal is certain if $B(\tau) > \lambda$
- Dots maximize feasible welfare
  - Higher λ raises the optimal tax



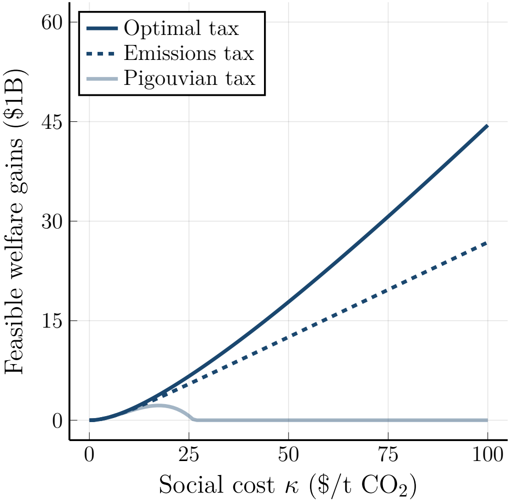
Optimal policy targets both emissions and value

- Pigouvian + shading
  - Pigouvian helps welfare gains
  - Shading helps repeal risk
- Avoids low-e, high-v munis (lighter)
  - Low welfare gains
  - High repeal risk
- Targets high-e, low-v munis (darker)
  - With continuous shading



Optimal policy avoids repeal

- Pigouvian tax collapses
  - When we account for repeal risk
  - High burden $\rightarrow$ inevitable repeal
- Emissions tax avoids repeal
  - Reduces tax rate for large κ
- Optimal tax survives and grows in κ

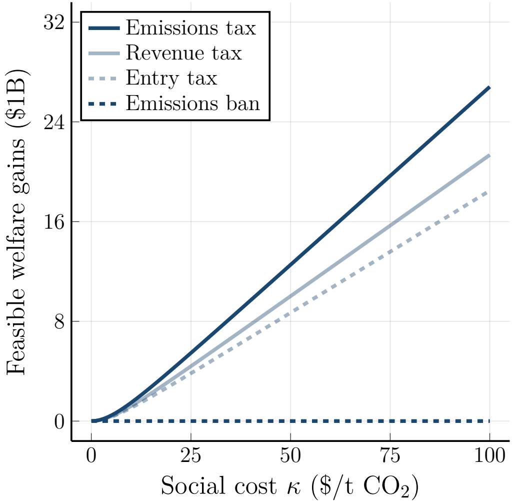


Practical policies capture substantial gains

- Emissions tax performs best
  - Targets observable carbon stocks
  - We have corr$(e, v) = -0.44$

- Revenue/entry taxes also good
  - Neither directly targets emissions

- But not municipal emissions bans
  - No revenue, high burden
  - No ban raises F over this range



Conclusion

Environmental regulation must be politically feasible

- In Brazil, conservation led to political opposition
  - Agricultural interests responded with money and votes
- Optimal policy balances abatement and burden